

Jesus Chavez: AI & Robotics

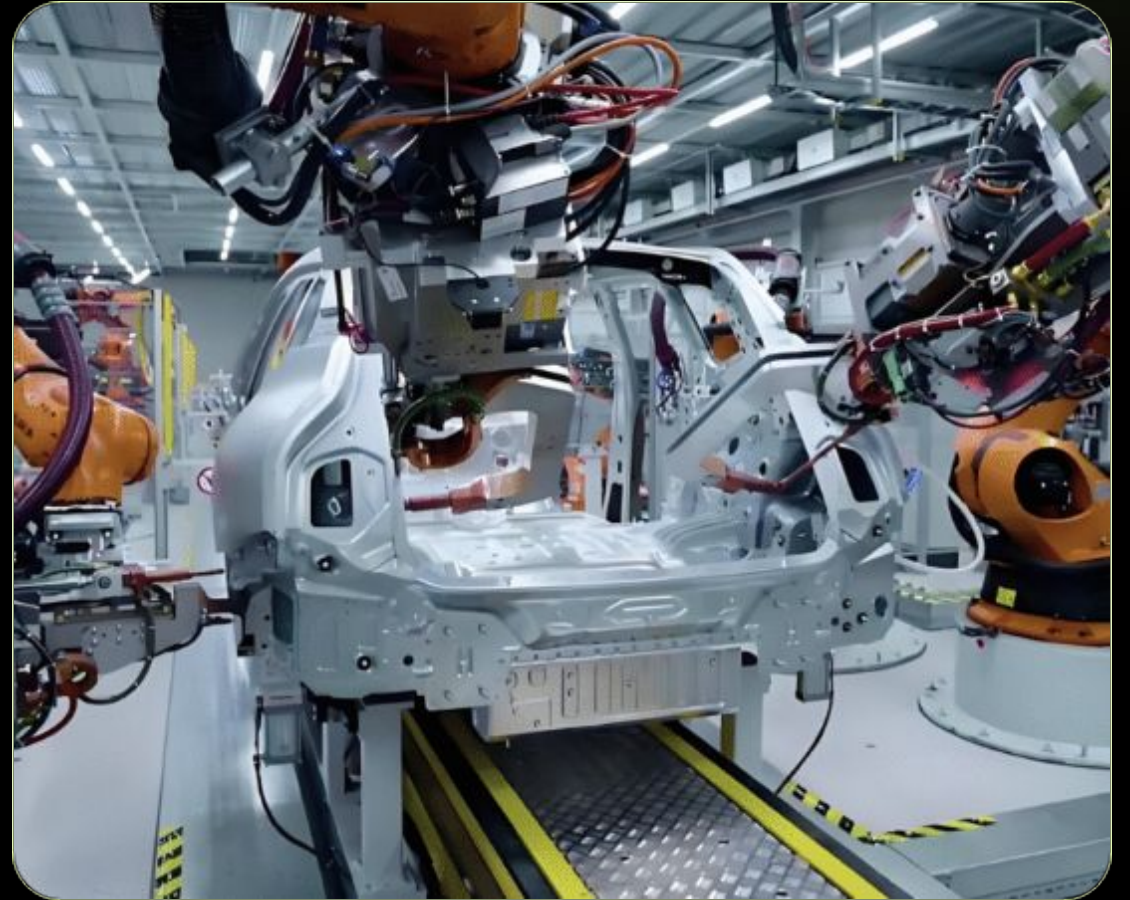
ITAI 2372 | Professional Portfolio

Expertise: Deep Learning Integration · Ethical Governance

| Industrial AI & Computer Vision

Cognitive Factory Implementation

- Mastered **Convolutional Neural Networks (CNNs)** for decentralized quality control in automotive assembly.
- Utilized **Synthetic Data Generation** to train models for rare defect detection with 90%+ accuracy.
- Implemented **Edge Computing** infrastructure to ensure zero-latency on-site diagnostics.



| Ethical AI & Deep Learning



Precision Medicine

Leveraged **U-Net architectures** for medical image segmentation and patient digital twins.



Ethical Governance

Applied **EU AI Act** risk frameworks to ensure transparency and prevent algorithmic bias.



Explainable AI

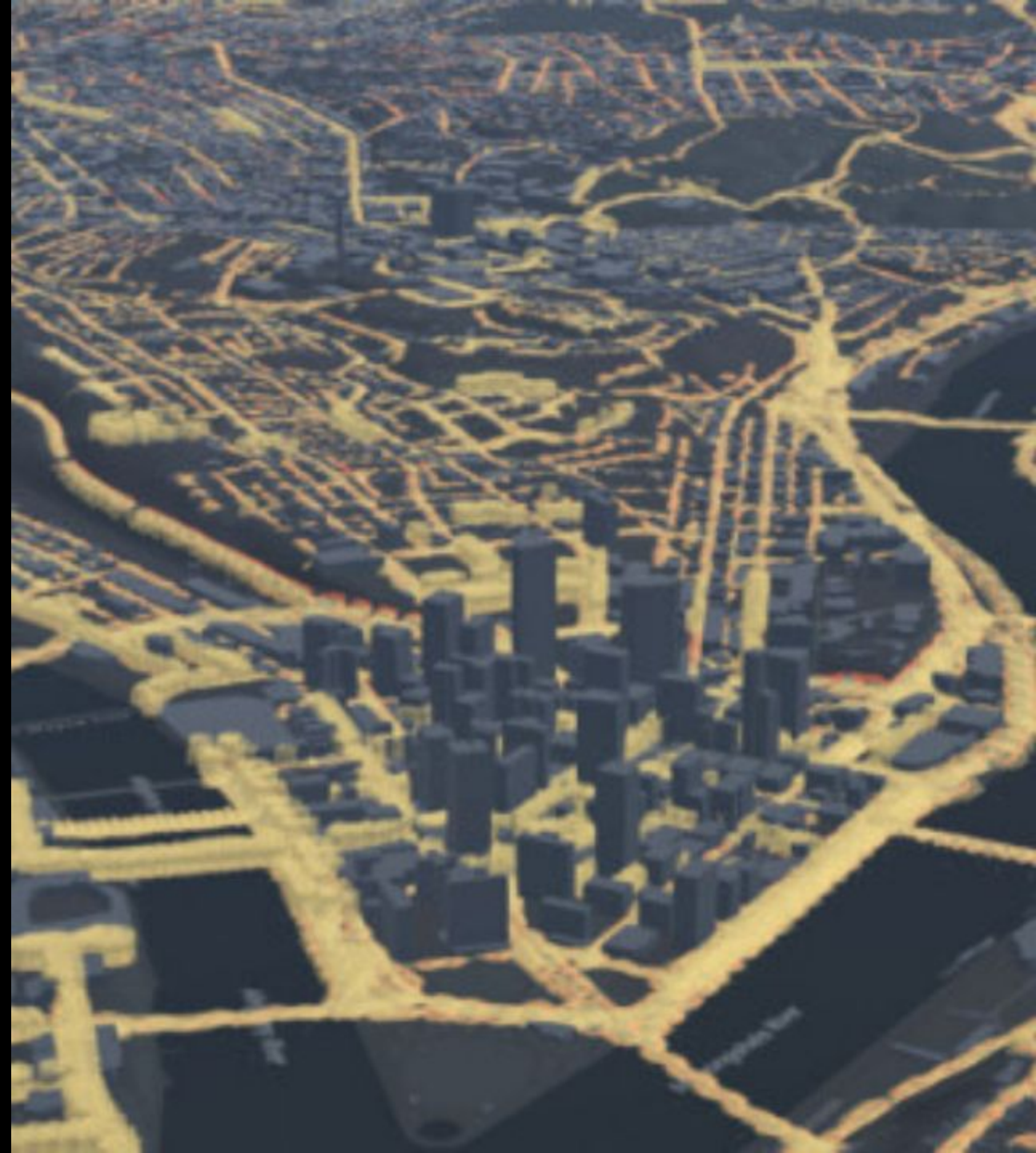
Solved the "Black Box" problem using **XAI** to provide clear decision rationale for clinicians.

ResilientGrid Solutions

Autonomous Infrastructure

Developing **Resilient Grid AI**: a decentralized network for real-time city management.

- **GNNs**: Modeling complex interdependencies between water and power grids.
- **IoT Fusion**: Autonomous emergency mitigation before human intervention.
- **Impact**: 30% reduction in urban service disruption during climate events.



Professional Credentials & GitHub

Core Competencies

Technical Skills: Python, PyTorch, ROS, Blockchain Integration, Generative Design (GANs/VAEs).

Industry Knowledge: Industry 5.0, HIPAA Compliance, Smart Nation Architectures.



https://github.com/J-xololt/JesusChavez_ITAI-2372_Portfolio



ITAI 2372: Artificial Intel Applications